QUESTION 1

Constraint-Based Problem Solving Using Backtracking Search

Problem: Assign D1, D2 and D3 to Morning (M), Afternoon (A) and Night (N).
Constraints: D1 ≠ N; D2 must be before D3; one doctor per shift; D3 ≠ M; each doctor gets exactly one shift.

Step-by-Step Calculation

Step 1: Try D1 = Morning.

Check	Calculation	Result
D1 cannot work Night	D1=M, so D1≠N	✓
Shift available	M is unused	✓
Decision	D1=M	ACCEPT

Current assignment: {D1=M}. Remaining shifts: A, N.

Step 2: Try D2 = Morning.

Check	Calculation	Result
One doctor per shift	M is already used by D1	✗
Decision	D2=M is invalid	REJECT → BACKTRACK

Step 3: Try D2 = Afternoon.

Check	Calculation	Result
Shift available	A is unused	✓
D2 before D3	D3 can still be assigned after A	✓
Decision	D2=A	ACCEPT

Current assignment: {D1=M, D2=A}. Remaining shift: N.

Step 4: Try D3 = Morning.

Check	Calculation	Result
D3 cannot work Morning	D3=M violates the constraint	✗
Decision	Reject this branch	REJECT → BACKTRACK

Step 5: Try D3 = Afternoon.

Check	Calculation	Result
One doctor per shift	A is already assigned to D2	✗
Decision	Reject this branch	REJECT → BACKTRACK

Step 6: Try D3 = Night.

Check	Calculation	Result
D3 cannot work Morning	D3=N, so satisfied	✓
D2 before D3	A < N	✓
One doctor per shift	N is unused	✓
Decision	D3=N	ACCEPT

Complete Search Trace

Step	Assignment	Result
1	D1=M	✓ Accept
2	D1=M, D2=M	✗ Reject
3	D1=M, D2=A	✓ Accept
4	D1=M, D2=A, D3=M	✗ Reject
5	D1=M, D2=A, D3=A	✗ Reject
6	D1=M, D2=A, D3=N	✓ Solution

Final Verification

Constraint	Verification	Status
D1 ≠ N	D1=M	✓
D2 before D3	A<N	✓
One doctor per shift	M, A, N used once	✓
D3 ≠ M	D3=N	✓
Exactly one shift each	D1=M, D2=A, D3=N	✓

Final Answer: Morning = D1, Afternoon = D2, Night = D3.

QUESTION 2

Robot Grid Using BFS Search and Manhattan Distance

The original question does not specify obstacle coordinates. The following 5×5 grid is therefore used to demonstrate the complete calculation.

	1	**2**	**3**	**4**	**5**
1	S	.	.	#	.
2	#	#	.	#	.
3	.	.	.	.	.
4	.	#	#	#	.
5	.	.	.	.	G

Start S=(1,1), Goal G=(5,5). Each legal move costs 1. Manhattan distance: h(n)=|row−5|+|column−5|.

BFS: Every Step

Step	**Current node**	**Movement checks**	**Queue after expansion**
0	(1,1)	R=(1,2)✓; D=(2,1); U/L outside	[(1,2)]
1	(1,2)	R=(1,3)✓; D=(2,2); L visited	[(1,3)]
2	(1,3)	D=(2,3)✓; R=(1,4); L visited	[(2,3)]
3	(2,3)	D=(3,3)✓; L/R blocked; U visited	[(3,3)]
4	(3,3)	L=(3,2)✓; R=(3,4)✓; D blocked	[(3,2),(3,4)]
5	(3,2)	L=(3,1)✓; U/D blocked	[(3,4),(3,1)]
6	(3,4)	R=(3,5)✓; U/D blocked	[(3,1),(3,5)]
7	(3,1)	D=(4,1)✓; U blocked; R visited	[(3,5),(4,1)]
8	(3,5)	U=(2,5)✓; D=(4,5)✓; L visited	[(4,1),(2,5),(4,5)]
9	(4,1)	D=(5,1)✓; R blocked; U visited	[(2,5),(4,5),(5,1)]
10	(2,5)	U=(1,5)✓; D/L blocked or visited	[(4,5),(5,1),(1,5)]
11	(4,5)	D=(5,5)=G✓	GOAL → STOP

Manhattan Distance + Path Cost

Node	g(n)	h(n) calculation	h(n)
(1,1)	0	$\|1-5\|+\|1-5\|$	8
(1,2)	1	$\|1-5\|+\|2-5\|$	7
(1,3)	2	$\|1-5\|+\|3-5\|$	6
(2,3)	3	$\|2-5\|+\|3-5\|$	5
(3,3)	4	$\|3-5\|+\|3-5\|$	4
(3,4)	5	$\|3-5\|+\|4-5\|$	3
(3,5)	6	$\|3-5\|+\|5-5\|$	2
(4,5)	7	$\|4-5\|+\|5-5\|$	1
(5,5)	8	$\|5-5\|+\|5-5\|$	0

Final path: S → (1,2) → (1,3) → (2,3) → (3,3) → (3,4) → (3,5) → (4,5) → G.

Number of moves = 8. Cost = 8 × 1 = 8. Manhattan distance from S = 8, so the obtained path reaches the theoretical minimum for four-direction movement.
Final Answer: Optimal path cost = 8.

QUESTION 3

Autonomous Rescue Robot – Uniform Cost Search and Online Search

The robot moves Up, Down, Left and Right. Obstacles cannot be entered. It sees adjacent cells only. Safe movement costs 1; a risky cell adds 2, so risky movement costs 1+2=3.

Step 1: Calculate Movement Costs

Move	Calculation	Cost
Safe	1+0	1
Risky	1+2	3
Obstacle	Not allowed	Reject

Because costs are different, the robot must compare accumulated costs rather than only counting steps.

Step 2: UCS Priority Queue Calculation

For the numerical example, A and C and G are safe; B is risky; D is safe.

Step	Expanded	Calculation	Priority queue
0	S	A=0+1=1; B=0+3=3	A(1), B(3)
1	A(1)	C=1+1=2	C(2), B(3)
2	C(2)	G=2+1=3	B(3), G(3)
3	B(3)	D=3+1=4	G(3), D(4)
4	G(3)	Goal reached	STOP

Step 3: Compare Complete Route Costs

Route	Calculation	Total
S→A→C→G	1+1+1	3
S→B→D→G	3+1+1	5
Difference	5−3	2

Since 3 < 5, the rational choice is S→A→C→G.

Step 4: Online Search / Knowledge Update

Step	Robot observes	Action / calculation
1	At S: A safe, B risky	A=1; B=3; choose A
2	At A: C discovered safe	C=1+1=2; choose C
3	At C: G discovered safe	G=2+1=3
4	G is lowest-cost goal	Move to G
5	If a new obstacle is discovered	Update map, reject blocked state, recalculate UCS

AI Concepts + Justification

Concept	Evidence
Intelligent agent	Sense → calculate → choose → move → sense again
Rationality	Chooses cost 3 instead of cost 5 because 3<5
Environment	Partially observable; robot sees only adjacent cells
Constraint analysis	U/D/L/R only; obstacle rejected; safe=1; risky=3
Why UCS	Different movement costs require accumulated-cost comparison
Why online search	Robot does not know the complete map initially

Final Answer: The worked UCS solution is S→A→C→G with a total cost of 3. If new information blocks a route, the robot updates its knowledge and repeats the cost-based search.